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جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم علم الحاسوب والمعلومات

Advanced Database

Assignment 2

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Task 1: Upload the Excel file into Oracle APEX and create a table named **DIRTY_BOOKS**.

Screenshot: Show the upload process and the created table **DIRTY_BOOKS**.

The screenshot shows the Oracle APEX SQL Workshop interface. The 'Object Browser' on the left lists various database objects, with 'DIRTY_BOOKS' highlighted under the 'Tables' section. The main panel displays the 'Data' tab for the 'DIRTY_BOOKS' table, showing a list of 12 rows. The columns are ID, BOOK_ID, BOOK_TITLE, AUTHOR, BORROWER_NAME, and BORROW_DATE. The data is as follows:

ID	BOOK_ID	BOOK_TITLE	AUTHOR	BORROWER_NAME	BORROW_DATE
1	1	the alchemist	paulo coelho	ali ahmed	1/1/2023
2	1	the alchemist	paulo coelho	ali ahmed	1/1/2023
3	2	Clean Code	robert martin	sara khalid	1/5/2023
4	3	deep work	cal newport	omar taha	1/7/2023
5	4	zero to one	peter thiel	omar taha	1/10/2023
6	5	problem solving	jacky rowling	nada yousif	1/12/2023
7	6	Atomic Habits	James Clear	ali ahmed	1/13/2023
8	7	sapiens	-	ammar fadi	1/15/2023
9	8	The Power	rhonda byrne	omar taha	1/15/2023
10	9	deep work	cal newport	omar taha	1/18/2023
11	10	the power	rhonda byrne	omar taha	1/18/2023
12	11	Clean Code	robert martin	sara khalid	1/16/2023

Task 2: Convert all text fields (Book_Title, Author, Borrower_Name) to **UPPER CASE** using **SQL UPDATE**.

Screenshot: Show the SQL command and the updated table **DIRTY_BOOKS**.

The screenshot shows the Oracle APEX SQL Workshop interface. The 'SQL' tab is active, displaying the following SQL command:

```
1 UPDATE DIRTY_BOOKS
2 SET BOOK_TITLE = UPPER(TRIM(BOOK_TITLE)),
3   AUTHOR = UPPER(TRIM(AUTHOR)),
4   BORROWER_NAME = UPPER(TRIM(BORROWER_NAME));
```

Below the SQL command, the 'Results' tab shows '12 row(s) updated.' Below that, the 'Results' tab displays the updated data for the 'DIRTY_BOOKS' table. The columns are ID, BOOK_ID, BOOK_TITLE, AUTHOR, BORROWER_NAME, and BORROW_DATE. The data is as follows:

ID	BOOK_ID	BOOK_TITLE	AUTHOR	BORROWER_NAME	BORROW_DATE
1	1	THE ALCHEMIST	PAULO COELHO	ALI AHMAD	1/1/2023
2	1	THE ALCHEMIST	PAULO COELHO	ALI AHMED	1/1/2023
3	2	CLEAN CODE	ROBERT MARTIN	SARA KHALID	1/5/2023
4	3	DEEP WORK	CAL NEWPORT	OMAR TAHA	1/7/2023
5	4	ZERO TO ONE	PETER THIEL	OMAR TAHA	1/10/2023
6	5	PROBLEM SOLVING	JACKY ROWLING	NADA YOUSIF	1/12/2023
7	6	ATOMIC HABITS	JAMES CLEAR	ALI AHMED	1/13/2023
8	7	SAPIENS	-	AMMAR FADI	1/15/2023
9	8	THE POWER	RHONDA BYRNE	OMAR TAHA	1/15/2023
10	9	DEEP WORK	CAL NEWPORT	OMAR TAHA	1/18/2023

Task 3: Fill missing values in Book_Title, Author, and Borrower_Name with 'UNKNOWN'.

Screenshot: Show the SQL commands and the updated table DIRTY_BOOKS after filling NULL values.

```
6 UPDATE DIRTY_BOOKS
7 SET BOOK_TITLE = 'UNKNOWN'
8 WHERE BOOK_TITLE IS NULL;
9
10 UPDATE DIRTY_BOOKS
11 SET AUTHOR = 'UNKNOWN'
12 WHERE AUTHOR IS NULL;
13
14 UPDATE DIRTY_BOOKS
15 SET BORROWER_NAME = 'UNKNOWN'
16 WHERE BORROWER_NAME IS NULL;
17
```

Results Explain Describe Saved SQL History

0 row(s) updated.

ID	BOOK_ID	BOOK_TITLE	AUTHOR	BORROWER_NAME	BORROW_DATE
1	1	THE ALCHEMIST	PAULO COELHO	ALI AHMAD	1/1/2023
2	1	THE ALCHEMIST	PAULO COELHO	ALI AHMED	1/1/2023
3	2	CLEAN CODE	ROBERT MARTIN	SARA KHALID	1/5/2023
4	3	DEEP WORK	CAL NEWPORT	OMAR TAHA	1/7/2023
5	4	ZERO TO ONE	PETER THIEL	OMAR TAHA	1/10/2023
6	5	PROBLEM SOLVING	JACKY ROWLING	NADA YOUSIF	1/12/2023
7	6	ATOMIC HABITS	JAMES CLEAR	ALI AHMED	1/15/2023
8	7	SAPIENS	UNKNOWN	AMMAR FADI	1/15/2023
9	8	THE POWER	RHONDA BYRNE	OMAR TAHA	1/15/2023
10	9	DEEP WORK	CAL NEWPORT	OMAR TAHA	1/18/2023

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds Download

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Task 4: Remove fully duplicated rows that have the same (Book_ID, Book_Title, Author, Borrower_Name, Borrow_Date) and keep only one row.

Screenshot: Show the SQL command for deleting duplicates and the final cleaned table DIRTY_BOOKS.

```
18 DELETE FROM DIRTY_BOOKS
19 WHERE ROWID NOT IN (
20     SELECT MIN(ROWID)
21     FROM DIRTY_BOOKS
22     GROUP BY BOOK_ID, BOOK_TITLE, AUTHOR, BORROWER_NAME, BORROW_DATE
23 );
24
25 select * from DIRTY_BOOKS;
```

Results Explain Describe Saved SQL History

0 row(s) deleted.

Task 5: Create a new table BOOKS (Book_ID, Book_Title, Author) and insert unique book data from DIRTY_BOOKS.

Screenshot: Show the SQL command for creating BOOKS, the insert command, and the final BOOKS table.

```
24
25 CREATE TABLE BOOKS (
26     BOOK_ID     NUMBER PRIMARY KEY,
27     BOOK_TITLE  VARCHAR2(100),
28     AUTHOR      VARCHAR2(100)
29 );
30
```

Results Explain Describe Saved SQL History

Table created.

```
30
31 INSERT INTO BOOKS (BOOK_ID, BOOK_TITLE, AUTHOR)
32 SELECT DISTINCT BOOK_ID, BOOK_TITLE, AUTHOR
33 FROM DIRTY_BOOKS;
```

Results Explain Describe Saved SQL History

11 row(s) inserted.

Task 6: Create a new table BORROW (Borrow_ID, Book_ID, Borrower_Name, Borrow_Date) with a foreign key referencing BOOKS. Insert borrow records from DIRTY_BOOKS into BORROW.

Screenshot: Show the SQL command for creating BORROW, the insert command, and the final BORROW table.

```
34
35 CREATE TABLE BORROW (
36     BORROW_ID     NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
37     BOOK_ID       NUMBER,
38     BORROWER_NAME VARCHAR2(100),
39     BORROW_DATE   DATE,
40     CONSTRAINT FK_BOOK FOREIGN KEY (BOOK_ID) REFERENCES BOOKS (BOOK_ID)
41 );

```

Results Explain Describe Saved SQL History

Table created.

```
42
43 INSERT INTO BORROW (BOOK_ID, BORROWER_NAME, BORROW_DATE)
44 SELECT BOOK_ID, BORROWER_NAME, BORROW_DATE
45 FROM DIRTY_BOOKS;
```

Results Explain Describe Saved SQL History

12 row(s) inserted.

Task 7: Display the final result by joining BOOKS and BORROW tables to show borrow records with book titles and authors.
Screenshot: Show the SQL JOIN command and the final result table.

```
47 SELECT
48   B.BORROW_ID,
49   BK.BOOK_TITLE,
50   BK.AUTHOR,
51   B.BORROWER_NAME,
52   B.BORROW_DATE
53 FROM BORROW B
54 JOIN BOOKS BK
55 ON B.BORROW_ID = BK.BORROW_ID;
```

BORROW_ID	BOOK_TITLE	AUTHOR	BORROWER_NAME	BORROW_DATE
1	THE ALCHEMIST	PAULO COELHO	ALI AHMAD	1/1/2023
2	THE ALCHEMIST	PAULO COELHO	ALI AHMED	1/1/2023
3	CLEAN CODE	ROBERT MARTIN	SARA KHALID	1/5/2023
4	DEEP WORK	CAL NEWPORT	OMAR TAHA	1/7/2023
5	ZERO TO ONE	PETER THIEL	OMAR TAHA	1/10/2023
6	PROBLEM SOLVING	JACKY ROWLING	NADA YOUSIF	1/12/2023
7	ATOMIC HABITS	JAMES CLEAR	ALI AHMED	1/15/2023
8	SAPIENS	UNKNOWN	AMMAR FADI	1/15/2023
9	THE POWER	RHONDA BYRNE	OMAR TAHA	1/15/2023
10	DEEP WORK	CAL NEWPORT	OMAR TAHA	1/18/2023

This query shows the borrow records with the corresponding book titles and authors.

Conclusion:

In this assignment, I successfully uploaded the dirty_books.xlsx file into Oracle APEX, cleaned the data by converting text fields to UPPER CASE, filled missing values, and removed duplicates.

I normalized the data by creating two separate tables: BOOKS and BORROW.

Finally, I inserted the cleaned and unique data into these tables and verified the results with SQL queries.

This process demonstrates the importance of data cleaning, normalization, and proper documentation in database management.

"واخر دعواهم ان الحمد لله رب العالمين"